a

Real Units

Unit 89 Unit 78 Unit 129 Unit 158 Unit 126

b

Idealized Units

Retinotopic Relational

r = 0.00 r = 0.25 r = 0.50 r = 0.75 r = 1.00

c

Monkey P (n = 998) Monkey E (n = 341)

89 78 129 158 126

Fraction of neurons

0.00 0.05 0.10 0.15 0.20

0.0 0.5 1.0

Relationality (r)

d

Monkey P Monkey E

Cumulative fraction

0.00 0.25 0.50 0.75 1.00

0.0 0.5 1.0

Relationality (r)

Weighted Avg.
Normalization
Inhibition
Basis Func.
Future Field
Data

e

Weighted Avg. Normalization Inhibition Basis Func. Future Field

f

θ_1 φ_2 φ_1 θ_2

$R(\theta_1, \varphi_1) > R(\theta_2, \varphi_1)$

$R(\theta_1, \varphi_2) < R(\theta_2, \varphi_2)$

g

Monkey P (n = 998) Monkey E (n = 341)

Relative likelihood

1.000 0.975 0.950 0.925 0.900

1.00 0.98 0.96

Weighted Avg. Normalization Inhibition Basis Func. Future Field

h

Receptive Field

Stimulus 1 (high firing rate)

Stimulus 2 (low firing rate)

Object
Receptive field
Future field
Response plan